1.1) $D = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & 1 \\ -1 & 4 & -1 & 0 \\ 0 & 2 & 1 & 1 \end{vmatrix} \overset{\substack{L_2-2L_1 \\ L_4+L_3}}{=} \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & -5 \\ 0 & 6 & -2 & 3 \\ 0 & 2 & 1 & 1 \end{vmatrix}$

$= 5 \begin{vmatrix} -1 & 1 & -1 \\ 6 & -2 & 3 \\ 2 & 1 & 1 \end{vmatrix} \overset{\substack{C_1+C_2 \\ C_3+C_2}}{=} \begin{vmatrix} 0 & 1 & 0 \\ 4 & -2 & 1 \\ 3 & 1 & 2 \end{vmatrix} = -5 \begin{vmatrix} 4 & 1 \\ 3 & 2 \end{vmatrix} = -25$

1.2) $\begin{vmatrix} -2-a & -1 & 1 \\ 5 & -1-a & 4 \\ 5 & 1 & 2-a \end{vmatrix} \overset{C_3+C_2}{=} \begin{vmatrix} -2-a & 0 & 1 \\ 5 & 3-a & 4 \\ 5 & 3-a & 2-a \end{vmatrix}$

$= (3-a) \begin{vmatrix} -2-a & 0 & 1 \\ 5 & 1 & 4 \\ 5 & 1 & 2-a \end{vmatrix} \overset{L_3-L_2}{=} (3-a) \begin{vmatrix} -2-a & 0 & 1 \\ 5 & 1 & 4 \\ 0 & 0 & -2-a \end{vmatrix}$

$= (3-a)(2+a)^2 = 0 \Rightarrow a_1 = 3$

$a_2 = -2$

1.3) $A = \begin{bmatrix} 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \overset{L_1 \leftarrow \frac{1}{2} L_1}{\sim} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \overset{L_3-2L_1}{\sim} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 & -1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$

$\overset{\substack{L_3-L_1 \\ L_2-L_1}}{\sim} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \overset{L_3 \leftarrow -\frac{1}{2} L_3}{\sim} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \operatorname{rank}(A) = 3$

1.4) $A = \begin{pmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{pmatrix}$

$$[A\,|\,I_3] = \left[\begin{array}{ccc|ccc} 2 & 2 & 3 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 2 & 1 & 0 & 0 & 1 \end{array}\right] \overset{L_1 \leftarrow -L_2}{\sim} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ 2 & 2 & 3 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\overset{\substack{L_2 - 2L_1 \\ L_3 + L_1}}{\sim} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 4 & 3 & 1 & -2 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{array}\right] \overset{L_2 \leftarrow -L_3}{\sim} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 4 & 3 & 1 & -2 & 0 \end{array}\right]$$

$$\overset{L_1 + L_2}{\sim} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 4 & 3 & 1 & -2 & 0 \end{array}\right] \overset{L_3 - 4L_2}{\sim} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 & -6 & -4 \end{array}\right]$$

$$\overset{\substack{L_3 + L_1 \\ L_3 + L_2}}{\sim} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -4 & -3 \\ 0 & 1 & 0 & 1 & -5 & -3 \\ 0 & 0 & -1 & 1 & -6 & -4 \end{array}\right] \overset{L_3 \leftarrow -L_3}{\sim} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -4 & -3 \\ 0 & 1 & 0 & 1 & -5 & -3 \\ 0 & 0 & 1 & -1 & 6 & 4 \end{array}\right]$$

$$\Rightarrow A^{-1} = \begin{pmatrix} 1 & -4 & -3 \\ 1 & -5 & -3 \\ -1 & 6 & 4 \end{pmatrix}$$

1.5) $\begin{cases} x + 2y + 4z = 4 \\ 5x + y + 2z = -7 \\ 3x - y + z = -6 \end{cases}$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 4 \\ 5 & 1 & 2 & -7 \\ 3 & -1 & 1 & 6 \end{array}\right] \overset{\substack{L_2 - 5L_1 \\ L_3 - 3L_1}}{\sim} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 4 \\ 0 & -9 & -18 & -27 \\ 0 & -7 & -11 & -6 \end{array}\right] \overset{L_2 \leftarrow -\frac{1}{9}L_2}{\sim} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & -7 & -11 & -6 \end{array}\right] \overset{L_3 + 7L_2}{\sim}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 3 & 15 \end{array}\right] \overset{L_3 \leftarrow \frac{1}{3}L_3}{\sim} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

$\Rightarrow z = 5 \Rightarrow y = -7 \Rightarrow x = -2$